

ARUNKUMAR K R

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🐙 GitHub | 🏆 Leetcode | 🏠 Hackerrank

OBJECTIVE

Computer Science and Engineering student passionate about creating efficient, scalable software solutions. Combines technical expertise in **modern web technologies** with a deep understanding of **AI/ML workflows**. Eager to join a dynamic team to solve real-world problems through clean code, robust architecture, and continuous innovation.

EDUCATION

KPR Institute of Engineering and Technology , Coimbatore	2023 – 2027
B.E. Computer Science and Engineering	CGPA: 8.78
Kongu Vellalar Matriculation Higher Secondary School , Perundurai	2022 – 2023
Higher Secondary Education	HSC: 94.6%

TECHNICAL SKILLS

- **Languages & Core CS:** Python, Java, C, SQL, JavaScript, DSA, OOP, DBMS
- **AI & Machine Learning:** Deep Learning (CNNs, Transformers), RAG Pipelines, LangChain, Computer Vision (YOLOv8, OpenCV), TensorFlow, PyTorch, Scikit-Learn, Pandas
- **Web Development:** React.js, Node.js, FastAPI, Flask, Streamlit, RESTful APIs, HTML5, CSS3
- **Databases & Cloud:** MongoDB, PostgreSQL, ChromaDB (Vector DB), Docker, Git/GitHub

EXPERIENCE

SmartED Innovations	<i>Machine Learning Intern</i>
May 2025 – July 2025	
• Designed and deployed scalable Machine Learning models as RESTful APIs using FastAPI, enabling real-time inference. • Containerized ML applications using Docker to ensure environment consistency and seamless deployment across production servers.	

PROJECTS

ProManage - Collaborative Project Management API	GitHub
<i>Tech Stack: Node.js, TypeScript, SQL</i>	
• Architected a scalable REST API with Role-Based Access Control (RBAC) and JWT authentication, securing endpoints for distinct user roles. • Optimized system performance using Redis caching and Node.js Streams for efficient large file handling, ensuring low-latency responses under load.	
RouteX - PDF Routing System	Live Demo GitHub
<i>Tech Stack: Python, PyMuPDF, OCR, NLP</i>	
• Developed an intelligent document routing system that classifies scanned PDFs using hybrid text extraction and OCR. • Integrated NLP techniques for automatic document summarization and department-wise classification.	
Just-Chat - AI Legal Assistant	Live Demo GitHub
<i>Tech Stack: Python, LangChain, ChromaDB, Llama 2</i>	
• Built a Retrieval-Augmented Generation (RAG) chatbot to answer complex legal queries from uploaded documents. • Semantic search implementation using vector embeddings reduced query response time by 40%.	

ACHIEVEMENTS & CERTIFICATIONS

- **Research Publication:** Published a paper titled *"Real-Time Surface Defect Detection in Metals using YOLOv8"* at the **3rd ICEAMST International Conference**.
- **DSA Problems:** Solved 400+ problems across various competitive coding platforms like Leetcode, Hackerrank
- Completed RedHat Certified System Administrator **RHCSA** Certification.
- Hackerrank Certifications: Problem Solving(Intermediate), Problem Solving(Basics), Python, Java, SQL(Basics)
- Completed **OpenCV Bootcamp** Certification offered by OpenCV University.
- **Executive Member** at Dept. association - served as coordinator for **Technical events**.